

Retrieval-Augmented Healthcare Question Answering Using seq2seq

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1. Problem Definition

1.1 Real-World Problem

People often search online when they experience symptoms such as headaches or dizziness. The large number of complex or unreliable results can make finding clear information difficult.

1.2 System Input and Output

Input: A healthcare question written by the user.

Output: A concise medical answer with its original source.

1.3 Why the Problem Matters

The system can provide quick and clear information from trusted medical sources. It could benefit the public, patients, and caregivers seeking understandable health information

1.4 Task Type

This is a **conditional sequence-to-sequence generation** task because the system generates an answer based on the user's question and retrieved medical information.

2. Dataset Description and Inspection

2.1 Dataset Source

Dataset name: MedQuAD: Medical Question Answering Dataset

Dataset access source: [lavita/MedQuAD on Hugging Face](#)

Dataset version: No formal numbered release is provided. The Hugging Face version contains one train split with 47,441 records.

Original repository: [Official MedQuAD repository](#)

2.2 Licensing and Access Conditions

License: Creative Commons Attribution 4.0 International (CC BY 4.0), as declared by the original MedQuAD repository.	Access conditions: Publicly available through Hugging Face and the original GitHub repository and appropriate credit must be given to the dataset authors.
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2.3 Dataset Size and Structure

The Hugging Face dataset contains one unsplit training collection with 47,441 records and 13 fields. It does not provide official training, validation and test sets.

2.3.1 Data summary

Shape: (47,441, 13)	Records with empty answers: 31,034	Records with non-empty answers: 16,407
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2.4 Targets and Class Balance

Target: Corresponding reference medical answer.	Target type: Variable-length text sequence.
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Class balance is not directly applicable because the model predicts a sequence of tokens rather than one categorical class.

2.5 data inspection

The question_type metadata reveals an uneven distribution of question categories in the cleaned dataset. The six most frequent question types are:

Information: 4,518	Symptoms: 2,747	Treatment: 2,440
Inheritance: 1,446	Frequency: 1,120	Genetic changes: 1,087

The following distributions will be inspected before training:

Input-question length	Target-answer length	Question-type frequency
Medical-source frequency	Percentage of targets exceeding the selected decoder length	

2.6 Planned train/validation/test split

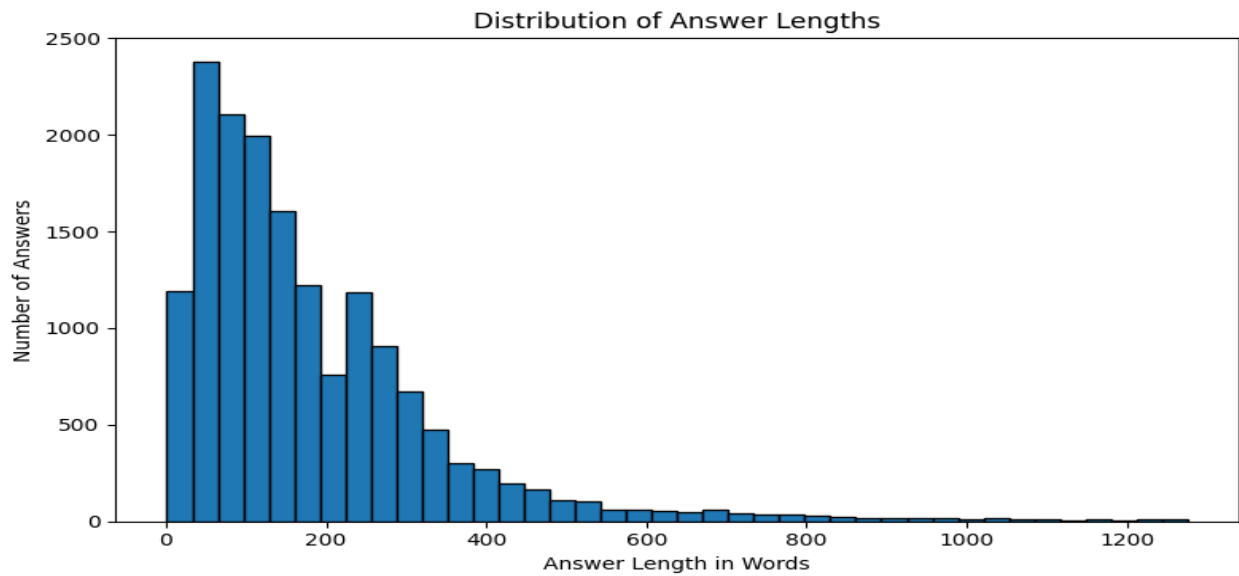
Training set: 13,027	Validation set: 1,650	Test set: 1,678
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2.7 Single Validation Metric

ROUGE-L will measure the similarity between generated and reference answers using their longest common subsequence. A higher score means greater similarity

3 Representative data samples table & Answer-length distribution figure

	question	question_type	document_source	answer
3634	What are the treatments for dihydropyrimidinase deficiency ?	treatment	GHR	These resources address the diagnosis or management of dihydropyrimidinase deficiency: - Genetic Testing Registry: Dihydropyrimidinase deficiency...
33249	Who is at risk for Parasites - Cysticercosis? ?	susceptibility	CDC	Cysticercosis is an infection caused by the larvae of the tapeworm, Taenia solium. A person with an adult tapeworm, which lives in the person's gu...
4395	What is (are) phenylketonuria ?	information	GHR	Phenylketonuria (commonly known as PKU) is an inherited disorder that increases the levels of a substance called phenylalanine in the blood. Pheny...
10277	Is Laron syndrome inherited ?	inheritance	GARD	Is Laron syndrome inherited? Most cases of Laron syndrome are inherited in an autosomal recessive manner. This means that to be affected, a person...
149	What are the treatments for globozoospermia ?	treatment	GHR	These resources address the diagnosis or management of globozoospermia: - Association for Reproductive Medicine: Semen Analysis - Centers for DI...



4. Repository and Supporting Code

[Github link](#)